Akash Gogate

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EDUCATION

University of Wisconsin at Madison*Graduating: May 2028*
Bachelor of Science in Computer Science and Biology ● Dean's List Fall 2024, Spring 2025, Fall 2025 ● 3.9 GPA
Coursework: Artificial Intelligence, Deep Learning, Advanced Algorithms, Data Structures, Cryptography, Linear Algebra, Machine Organization and Programming, Operating Systems, AP Computer Science A, AP Computer Science Principles, AP Statistics, Computer Science Capstone, Game Development, Web Development

TECHNOLOGIES

Languages: Python, Java, TypeScript, Node.js, R, Rust, JavaScript, C#, C++, SQL, NoSQL, Linux/Bash, Shell Scripting
ML & AI: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, Sentence-Transformers, Hugging Face, Transformer Architectures, Neural Networks, Deep Learning, NLP, Generative AI, Random Forest, Feature Engineering, Hyperparameter Tuning, Model Fine-Tuning, Model Deployment, Inference Optimization, GPU-Accelerated Computing, CUDA, Biostatistics, Data Visualization
Agentic & LLM: LLM Agents, LLM APIs, GPT-4o, RAG, Semantic Similarity Search, Vector Databases, MLflow, Weights & Biases, A/B Testing, Benchmarking, Experiment Tracking, Model Monitoring, Model Serving, Inference Serving
Backend & Systems: Kafka, Kubernetes, Docker, MongoDB, SQLite, Redis, Supabase, REST APIs, GraphQL, FastAPI, Microservices, Distributed Systems, System Design, Fault-Tolerant Systems, Low-Latency Systems, Scalable Architecture, CI/CD, Event-Driven Architecture
Frontend & Tools: React.js, Android SDK, Spring Boot, HTML/CSS, GitHub Actions, Git, Linux, Version Control, pytest, Unit Testing, Agile, Object-Oriented Design, Jupyter Notebook, Cross-Functional Collaboration

EXPERIENCE

The Kendziorski Lab at the University of Wisconsin-Madison*September 2025 - Present*
Machine Learning Research Intern
- Built end-to-end ML pipeline for scRNA-seq + spatial transcriptomics using PyTorch/TensorFlow deep learning; agentic framework selected and evaluated models across scVI/scANVI/scGen/AmortizedLDA for scalable production genomics, **reducing GPU computation 300%**; tracked via Weights & Biases
- Engineered TransferAgent automating natural language processing (NLP) and semantic similarity literature synthesis and cross-paper validity scoring; cut review time by **~60%**; collaborated with clinicians on clinical genomics impact
- Validated model performance and pipeline reproducibility across interdisciplinary research team; findings support ongoing clinical genomics publication, drug discovery, and precision medicine workflows

Leidos*May 2025 - Present*
Software Engineer Intern
- Optimized multi-objective satellite scheduling via memoization + dynamic programming in Python, **reducing constraint violations** under strict real-time operational constraints
- Designed scalable Kafka event-driven system design pipeline to MongoDB with **at-least-once delivery, fault-tolerant streaming**, and full system decoupling for mission-critical satellite ops
- Deployed containerized Docker test suite on Kubernetes via **GitHub Actions CI/CD**; comprehensive unit testing and debugging coverage ensuring production reliability and software engineering quality across all system components

Inspirit AI Research Internship*Sep 2022 - Mar 2023*
Machine Learning Research Intern
- Built skin cancer detection pipeline using OpenCV preprocessing and scikit-learn classifiers achieving **93% classification accuracy**; published as peer-reviewed paper "Early Skin Cancer Detection Improvement" through Inspirit AI
- Separately trained a Random Forest classifier on miRNA sequences achieving **95% predictive accuracy ($p < 0.05$)** for cancer cell likelihood prediction, validated via stratified k-fold cross-validation across multiple data splits

TECHNICAL PROJECTS

Self-Improving LLM Agent (https://github.com/AkashGogate/SelfImprovingLLMAgent)*Jan 2026 - May 2026*
- Architected failure-driven self-improvement loop for LLM agents across **5,000+ task benchmark**, consistently outperforming ReAct baseline by **15pp** via RAG-enhanced retrieval, Transformer-based reasoning, and failure-case feedback loops; engineered custom evaluation infrastructure with LLM-as-judge harness and novel repeated_failure_rate metric

Lotus Health (https://github.com/AkashGogate/lotus-health)*March 2026*
- Built clinical risk scoring engine across **45M insurance records** on **1,080-node ICD-10 graph**; integrates with **HIPAA-compliant, EHR-adjacent** clinical decision support workflows as validated patient-level risk score
- RAG pipeline with generative AI (Groq Llama) matches patient symptoms to ICD-10 disease risk profiles via SQL-queried healthcare and insurance data; all clinical outputs **hallucination-free and auditable**

LLM-Based SAT Practice Test Generator (https://github.com/AkashGogate/SATPracticeTestGenerator)*Jun 2024*
- Built full-stack SAT prep app (TypeScript/JavaScript/React.js + Node.js/FastAPI); integrated GPT-4o generative AI for adaptive question generation via REST API with session state, scoring, configurable difficulty; agile development with version control

miRcore Cancer Detection Research*Jul 2021*
- Trained and validated Random Forest classifier in R on miRNA sequences achieving **95% predictive accuracy ($p < 0.05$)**; validated via stratified k-fold cross-validation for cancer cell likelihood prediction

LEADERSHIP

Princeton Racket Club*May 2024 - Aug 2024*
Tennis Coach and Tournament Director
- Directed tournament logistics and player registration; coached athletes of all skill levels; managed client communications

Tennis Racket Stringing Services*Jan 2019 - Present*
- Founded and scaled to **50+ loyal clients**; manage all operations, inventory, and client consultations